SANTA CLARA UNIVERSITY
Leavey School of Business

Natural Language Processing and AI

ISBA 2411 | Weekly Learning Outcomes

Santa Clara University | *Leavey School of Business*

Weekly Learning Outcomes

Outcomes are aligned to Bloom's Taxonomy (Remember → Understand → Apply → Analyze → Evaluate → Create).

Week 1: NLP as a Socio-Technical System

Key Topics: Ambiguity and NLP task landscape; tokenization & normalization; language models vs LLMs; transformer ecosystem; foundation models in organizations; risk & governance

Learning Outcomes:

- Explain why natural language is inherently ambiguous and how this affects NLP system design
- Differentiate between traditional language models and large language models
- Identify major NLP task categories and their real-world applications
- Describe NLP systems as socio-technical artifacts with organizational, ethical, and governance implications

Week 2: Language as Structured Data

Key Topics: Subword tokenization; POS & NER; constituency parsing; syntactic ambiguity; embeddings as linguistic interfaces; tokenizer–model mismatch; multilingual pipelines

Learning Outcomes:

- Apply tokenization and normalization techniques to raw text corpora
- Interpret linguistic annotations such as POS tags and named entities
- Analyze sources of syntactic ambiguity and their downstream effects
- Evaluate tokenizer choices for multilingual and pretrained models

Week 3: Supervised Learning for NLP

Key Topics: Text classification; logistic regression; term-document matrices; feature engineering; FastText, spaCy & GLMNet; multimodal classification (AutoGluon); classifier evaluation

Learning Outcomes:

- Build supervised text classification pipelines using logistic regression, FastText, and spaCy
- Represent documents with term-document matrices and engineered features
- Evaluate classifiers using appropriate metrics and error analysis
- Apply multimodal classification approaches to applied business problems

Week 4: Text Retrieval & Deep Learning Intro

Key Topics: Web scraping & data collection; sentiment analysis; summarization; document similarity & retrieval; introduction to neural networks; PyTorch

Learning Outcomes:

- Collect and preprocess text data, including web scraping, for NLP tasks

- Apply sentiment analysis and summarization to real-world text
- Measure document similarity and build basic retrieval pipelines
- Explain neural network fundamentals and implement simple models in PyTorch

Week 5: Deep Learning & Topic Modeling

Key Topics: Deep learning (continued); Hugging Face pipelines for sentiment & QA; topic modeling; clustering; dimensionality reduction (t-SNE/UMAP)

Learning Outcomes:

- Use Hugging Face pipelines for sentiment analysis and question answering
- Apply topic modeling and clustering to uncover structure in text corpora
- Use dimensionality reduction to visualize high-dimensional representations
- Connect deep learning foundations to transformer-based approaches

Week 6: Embeddings & Midterm

Key Topics: Word & document embeddings; embedding visualization (t-SNE); economic & financial text analysis; FOMC transcripts; midterm review

Learning Outcomes:

- Generate and apply word and document embeddings
- Visualize and interpret embedding spaces using t-SNE
- Analyze domain text such as economic news and FOMC transcripts with embeddings
- Consolidate and demonstrate mastery of Weeks 1–5 concepts (midterm)

Week 7: Language Models & Transformers

Key Topics: Transformer architecture & self-attention; positional encoding; BERT & RoBERTa; masked language modeling; pretraining & fine-tuning; Hugging Face Transformers

Learning Outcomes:

- Explain the transformer architecture and the self-attention mechanism
- Describe how BERT and RoBERTa are pretrained (masked LM) and fine-tuned
- Load, fine-tune, and apply transformer models with Hugging Face
- Compare encoder, decoder, and encoder–decoder model designs

Week 8: LMs, Generative AI & RAG

Key Topics: Natural language generation; retrieval-augmented generation (RAG); LangChain & LangGraph; agentic workflows; multimodal LMs; interactive prototypes (Streamlit/Shiny)

Learning Outcomes:

- Explain the motivation and architecture of retrieval-augmented generation
- Build RAG pipelines and orchestrate workflows with LangChain and LangGraph
- Apply generative and multimodal models to applied tasks
- Build interactive prototypes and evaluate grounding and hallucination risks

Week 9: Evaluation, Ethics & Responsible AI

Key Topics: LLM evaluation & benchmarks; RLHF & alignment; reinforcement learning basics; fairness & bias; privacy; explainability (Shapley values); red-teaming

Learning Outcomes:

- Evaluate LLMs using benchmarks and task-appropriate metrics
- Analyze bias, fairness, and privacy risks in NLP systems

- Explain RLHF and alignment, and apply interpretability tools such as Shapley values
- Conduct structured red-teaming and risk assessments

Week 10: Final Presentations & Operationalizing NLP

Key Topics: Team final presentations & demos (Aug 18 & 20); LLMOps; monitoring & drift; governance & privacy; deployment architectures; build-vs-buy; technical storytelling & ROI framing; course wrap-up

Learning Outcomes:

- Design end-to-end NLP system architectures for production use
- Apply monitoring, drift detection, governance, and privacy requirements
- Communicate NLP system value, ROI, and risk to non-technical stakeholders
- Present an end-to-end NLP system with clear articulation of risks and benefits